

Multiplicity of traveling waves to bistable problems with negative diffusivity

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Joint work with

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Diffusion-convection reaction equations

We deal with the following parabolic equation in one spatial dimension:

$$\rho_t + f(\rho)_x = \{D(\rho) \rho_x\}_x + g(\rho) \quad \text{for } x \in \mathbb{R}, t \geq 0. \quad (1)$$

- ▶ The unknown is $\rho = \rho(x, t) \in [0, 1]$. (In view of applications ρ stands for a (normalized) **density**.)
- ▶ Equation (1) arises as a model in several applications where a **reaction** g is propagated in the space both in a **convective** f and **diffusive** D way.

Main interesting feature: D can be **negative**. This occurs in the context of, e.g.:

- ▶ *collective movements* (vehicular flows and crowd dynamics);
- ▶ several *physical* and *biological* models.

$D < 0$ near to 1, the negativity of D simulates an **aggregative behavior under high density of the agents**:

- ▶ high car densities and limited sight distance ahead in *vehicular flows* (Nelson (2000));
- ▶ panic behavior in *overcrowded dynamics* (Colombo-Rosini (2005) or Corli-Malaguti (2019)).

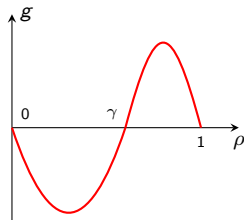
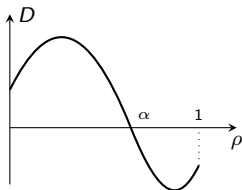
Main assumptions on the constitutive functions

(f) $f \in C^1([0, 1])$ with $f(0) = 0$;

(g) $g \in C^0([0, 1])$, $g(0) = g(1) = 0$ and **bistable** (Allen-Cahn):

(D) $D \in C^1([0, 1])$ that **changes sign** in $(0, 1)$;

- Equation (1) is a **forward** parabolic equation where $D > 0$ and a **backward** parabolic equation where $D < 0$. (D can vanish at the extrema of $[0, 1]$.)



(D_{pn}): from positive to negative

Our focus: wavefronts

- **Wavefront** solutions are traveling-wave, so that

$$\rho(x, t) = \varphi(x - ct),$$

for some **wave profile** $\varphi = \varphi(\xi)$ and **wave speed** $c \in \mathbb{R}$, such that the profile φ is *globally-defined, non-constant and monotone (decreasing)*.

- Under these assumptions, a wavefront ρ satisfies (1) if and only if its profile $\varphi : \mathbb{R} \rightarrow [0, 1]$ satisfies

$$\left(D(\varphi) \varphi' \right)' + (c - h(\varphi)) \varphi' + g(\varphi) = 0, \quad (2)$$

where $h = \dot{f}$, and

$$\varphi(-\infty) = 1 \quad \text{and} \quad \varphi(\infty) = 0. \quad (3)$$

If a profile satisfying (2) - (3) exists, we say that $c \in \mathbb{R}$ is an **admissible speed** for Equation (1).

- **Remark:** Formally, in order to seek wavefronts, we ask that φ is continuous, a.e. differentiable, $D(\varphi)\varphi' \in L^1_{\text{loc}}$ and φ solves (2) in **distributional sense**.
- **Remark:** Solutions of (2)-(3) (and solutions of (1)) are always **determined up to shifts**: If φ is a solution with speed c then, for every $x_0 \in \mathbb{R}$, the map $x \mapsto \varphi(x - x_0)$ is still a solution of (2)-(3) with the same speed c .

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Our contribution

In order to understand the **interpaly between convective and diffusive flows** and to extend and make precise some previous results in a unique framework, in collaboration with Corli (Ferrara) and Malaguti (Modena and Reggio Emilia), we have provided the existence, regularity and (strictly) monotonicity of wavefronts for Equation (1) under f :

- ▶ (D_{pn}) , (g) and (f) :
[B.-Corli-Malaguti, *Diffusion-convection reaction equations with sign-changing diffusivity and bistable reaction term*, submitted 2021].

Other contributions are contained in

- ▶ (D) , g monostable and (f) :
[B.-Corli-Malaguti, *Traveling waves for degenerate diffusion-convection reaction equations with sign-changing diffusion*, Discrete Contin. Dyn. Syst., 2021].
- ▶ Some preliminary results involving $D > 0$ and $g > 0$ and (f) :
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Our contribution

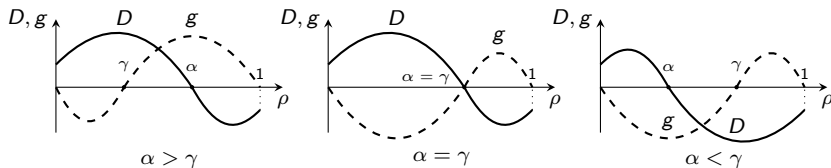
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Focus on (D_{pn}) and g bistable. Here, it results discriminant the mutual position of the (internal) zeros of D and g , respectively.



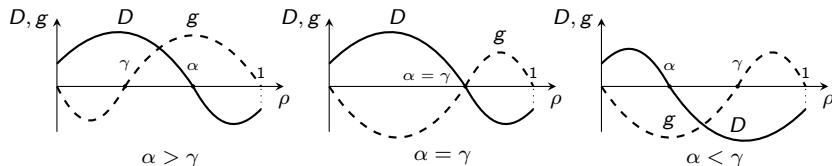
The context in which we move

Positive D and bistable g : profiles exist in correspondence of a unique speed c^* .

- ▶ $D = 1$ and $f = 0$: Aronson-Weinberger or Fife-McLeod in the '70s.
- ▶ $D > 0$ and f nonlinear: Malaguti-Marcelli-Matucci (2004).
- ▶ Further results under the bistable condition can be found in Hamel (2016).

Sign-changing D and bistable g with no convection

- ▶ $f = 0$ and (D_{pn}) : Maini-Malaguti-Marcelli-Matucci (2007).
- ▶ $f = 0$ and (D_{pnp}) : Kuzmin-Ruggerini (2011).



Main results

Theorem 1 (Admissible speeds, [B.-Corli-Malaguti (submitted 2021)])

Let $\mathcal{J}$ be the set of admissible speeds. Then either $\mathcal{J}$ is empty or

(a) $\alpha > \gamma$: then

$\mathcal{J}$ contains a unique value

▷ the corresponding profile is unique (up to shifts);

(b) $\alpha = \gamma$: then

$\mathcal{J}$ is a bounded interval

▷ for $c \in \mathcal{J}$ the profile is unique up to shifts and *stretchings* at level γ .

▷ Moreover, if it holds

$$|g(\varphi)| \geq L|\varphi - \gamma|^\tau \quad \tau \in (0, 1), \quad \text{in a neighborhood of } \gamma, \quad (\heartsuit)$$

then $\mathcal{J}$ is also closed;

(c) $\alpha < \gamma$: then

$\mathcal{J}$ is a bounded interval

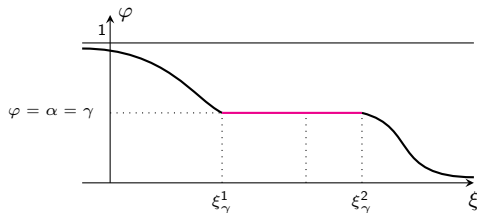
▷ for $c \in \text{int} \mathcal{J}$ there is a family of profiles $\{\varphi_\lambda\}_\lambda$;

▷ Every φ_λ is unique (up to shifts) such that $\varphi'_\lambda(\xi_\gamma) = \lambda \in [\lambda_c, 0)$;

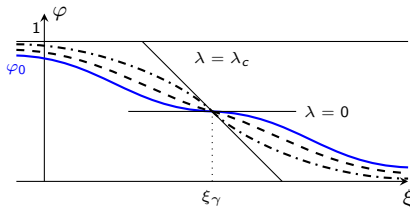
▷ If $(\heartsuit)$ holds, then $\mathcal{J}$ is also closed and for $c \in \mathcal{J}$ there exist profiles corresponding to $\lambda = 0$; φ_0 is unique up to shifts and *stretchings* at level γ .

Comments on Theorem 1

- ▶ If $\alpha > \gamma$ we generalize the case $f = 0$ (Maini-Malaguti-Marcelli-Matucci (2007));
- ▶ If $\alpha \leq \gamma$, **in contrast** with the case $f = 0$, there can exist admissible speeds (even non-unique);
- ▶ If $\alpha = \gamma$ uniqueness is up to shifts and **stretchings at level γ** :



- ▶ if $\alpha < \gamma$ profiles are **nonunique** (cone of admissible slopes; below profiles are shifted so that $\varphi(\xi_\gamma) = \gamma$), and φ_0 can be stretched:



Some more comments on Theorem 1

- ▶ In (b) and (c) of Theorem 1 the set $\mathcal{J}$ can degenerate to a single value.
- ▶ We also show that $\varphi' < 0$, with the possible exceptions of the points where φ reaches $0, \alpha, \gamma, 1$.
- ▶ Precise estimates on $\min \mathcal{J}$ and $\max \mathcal{J}$ can be given, as application of [BCM1]¹ (and reference therein).

¹[B.-Corli-Malaguti, *Electron. J. Qual. Theory Diff. Equ.* (2020)]

Main tool: reduction to singular first-order problems

- ▶ The **lack of regularity** properties of the involved functions $\implies$ it is **not** possible to apply a standard dynamical system analysis, in order to solve (2)-(3).
- ▶ Due to the monotonicity of φ , we instead take advantage of a method well-formalized by **Marcelli-Malaguti** in the early '00s: **reduction** (in regions where $D \neq 0$) of (2)-(3) into a singular first-order boundary problem for the function

$$z(\varphi) := D(\varphi)\varphi', \quad (4)$$

where the r.h.s. is evaluated at $\varphi^{-1}(\varphi)$.

- ▶ $\alpha > \gamma$: there exists a one-to-one correspondence between profiles φ (up to space shifts) with speed c and solutions z of

$$\begin{cases} \dot{z}(\varphi) = h(\varphi) - c - \frac{D(\varphi)g(\varphi)}{z(\varphi)} & \text{for } \varphi \in (0, \alpha) \cup (\alpha, 1), \\ z(\varphi) < 0 & \text{for } \varphi \in (0, \alpha), \\ z(\varphi) > 0 & \text{for } \varphi \in (\alpha, 1), \\ z(0) = z(\alpha) = z(1) = 0. \end{cases} \quad (5)$$

- ▶ $\alpha = \gamma$: profiles imply the solvability of (5), again.

- ▶ $\alpha < \gamma$: If g is regular enough (near γ) then there exists a one-to-one correspondence between profiles φ (up to space shifts) with speed c and solutions z of (5).

Otherwise (if $(\heartsuit)$ holds) the existence of the profiles implies only the solvability of

$$\begin{cases} \dot{z}(\varphi) = h(\varphi) - c - \frac{D(\varphi)g(\varphi)}{z(\varphi)} & \text{for } \varphi \in (0, \alpha) \cup (\alpha, \gamma) \cup (\gamma, 1), \\ z(\varphi) < 0 & \text{for } \varphi \in (0, \alpha), \\ z(\varphi) > 0 & \text{for } \varphi \in (\alpha, \gamma) \cup (\gamma, 1), \\ z(0) = z(\alpha) = z(1) = 0, \end{cases} \quad (6)$$

where $z(\gamma)$ can vanishes.

- ▶ A fine study of the first-order problems, by means of a **refined use of upper- and lower-solutions**, gives the results for (1).
- ▶ In particular, for any fixed $c \in \mathbb{R}$:
 - ▶ if $\alpha \geq \gamma$: Problem (5) has (at most) a unique solution z ;
 - ▶ if $\alpha < \gamma$: Problems (5) and (6) can admit a family of solutions, parameterized by $z(\gamma)$.

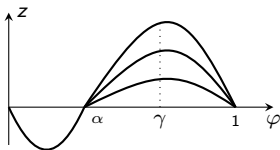


Figure: The multiple solutions z in case $\alpha \leq \gamma$.

Convex convective term

The next theorem provides necessary and sufficient conditions for the existence of profiles when f is convex.

Theorem 2 ([B.-Corli-Malaguti (submitted 2021)])

Assume that f is convex.

(Necessary condition) If Equation (1) has wavefronts then

$$\int_0^\alpha D(\rho)g(\rho) d\rho > 0. \quad (\star)$$

(Sufficient condition) Equation (1) has wavefronts if

$$\int_0^\alpha D(\rho)g(\rho) d\rho \geq \frac{2\alpha^2\Sigma}{3} \left(\Sigma + \sqrt{\Sigma^2 + \frac{2}{\alpha}M} \right), \quad (\star\star)$$

where

$$\Sigma := h(1) - h(0) + 2 \sqrt{\sup_{[\alpha,1]} \delta(Dg,1)} \quad \text{and} \quad M := \max_{\rho \in [0,\gamma]} -D(\rho)g(\rho) > 0,$$

Comments on Theorem 2

- ▶ We generalize the case $f = 0$.
- ▶ $(\star)$ (i.e. $\int_0^\alpha D(\sigma)g(\sigma) d\sigma > 0$) $\implies \alpha > \gamma$ (since otherwise $Dg < 0$ in $(0, \alpha)$) $\implies c > h(\alpha)$.
- ▶ Between $(\star)$ and $(\star\star)$ there is a **gap**, yet in the case $f = 0$.
- ▶ $(\star) \implies$ the region where $Dg > 0$ must prevail on the region where $Dg < 0$; the r.h.s. in $(\star\star)$ involves only regions where $Dg < 0$;
- ▶ Condition $(\star)$ does not depend on f , but this is a mere consequence of the assumption of convexity.

Sufficient conditions to have wavefronts in all cases - strictly concave convective term

The next result concerns some sufficient conditions for the existence of the profiles in the case f is strictly concave

Theorem 3 ([B.-Corli-Malaguti (submitted 2021)])

Assume that f is strictly concave. Then, Equation (1) admits wavefronts if

(a) $\alpha > \gamma$, $\dot{g}(\gamma) > 0$ and

$$\dot{f}(\gamma) - \frac{f(1) - f(\alpha)}{1 - \alpha} \geq 2 \left(\sqrt{\sup_{[\alpha, 1)} \delta(Dg, 1)} + \sqrt{\sup_{[0, \gamma)} \delta(Dg, \gamma)} \right); \quad (*)$$

(b) $\alpha = \gamma$, $(\heartsuit)$ and $(*)$;

(c) $\alpha < \gamma$ and

$$\dot{f}(\gamma) - \frac{f(1) - f(\gamma)}{1 - \gamma} > 2 \left(\sqrt{\sup_{[\gamma, 1)} \delta(Dg, 1)} + \sqrt{\sup_{[0, \gamma] \setminus \{\alpha\}} \delta(Dg, \alpha)} \right).$$

Notation: $\delta(Dg, 1)$ is the difference quotient of the function Dg w.r.t. a point 1:

$$\delta(Dg, 1)(s) := \frac{(Dg)(1) - (Dg)(s)}{1 - s} = \frac{D(s)g(s)}{s - 1} > 0.$$

Comments on Theorem 3

- ▶ Conditions (a)-(b)-(c) depend on the fine estimates that we can provide on the sub-thresholds of the problem and the characterization in terms of mutual relation between them of the solvability of the first-order singular problems.
- ▶ In order to fulfill (*), a necessary assumption is that Dg has finite slope at γ^- . Thus, in (a) the condition ($\heartsuit$) is implicitly excluded ($D(\gamma) \neq 0$).
- ▶ In (b), instead, conditions ($\heartsuit$) and (*) are not in contradiction because, in this case, Dg has zero derivative at γ .
- ▶ For any fixed D and g (regular enough), if f is sufficiently strictly concave then it satisfies the proper conditions in Theorem 3.
- ▶ The convexity assumption here has the role to better present the sufficient conditions but they can be provided more generally for a non-convex f .

Further results

- ▶ The behavior of φ where D degenerates can be characterized.
 - ▶ In contrast with the case $f = 0$, when profiles are always classical at α , according to some implicit conditions on the wave speed it can be possible to show **the occurrence of sharp profiles at α** , i.e. profiles with infinite slope or with different right- and left-derivatives when they reach α . (See also [BCM2].)
 - ▶ As well as the regularity of the wavefronts approaching 0 and 1 and information on their strict monotonicity can be deduced.
 - ▶ The other cases, i.e.:
 - ▶ g counter-bistable (first positive then negative);
 - ▶ D satisfies (D_{pn}) and g is either bistable and counter-bistable;
 - ▶ D changes sign more than once;
- are currently in preparation [B. (in preparation 2021)].

Thanks for your attention!

A formal definition

We define now traveling waves and the related profiles. We denote with $I \subseteq \mathbb{R}$ an open interval. We refer to [Gilding-Kersner \(2004\)](#).

Definition of TWs

Let c be a real constant and φ be a function such that

- $\varphi \in C(I)$ with values in $[0, 1]$, for some $I \subset \mathbb{R}$;
- φ is differentiable a.e.;
- $D(\varphi)\varphi' \in L^1_{\text{loc}}(I)$.

The function $\rho(x, t) := \varphi(x - ct)$, for (x, t) with $x - ct \in I$, is a **traveling-wave** solution of equation (1) with wave speed c and wave profile φ if, for every $\psi \in C_0^\infty(I)$,

$$\int_I \left(D(\varphi(\xi)) \varphi'(\xi) - f(\varphi(\xi)) + c\varphi(\xi) \right) \psi'(\xi) - g(\varphi(\xi)) \psi(\xi) d\xi = 0.$$

A TW solution is

global if $I = \mathbb{R}$ and **strict** if $I \neq \mathbb{R}$ and φ is not extendible to $\mathbb{R}$;

classical if φ is differentiable, $D(\varphi)\varphi'$ is absolutely continuous and (2) holds a.e.;

sharp at ℓ if there exists $\xi_\ell \in I$ such that $\varphi(\xi_\ell) = \ell$, with φ classical in $I \setminus \{\xi_\ell\}$ and not differentiable at ξ_ℓ ;

a **wavefront** if it is global, with a monotonic, non-constant profile φ satisfying either (3) or the converse condition;

a **semi-wavefront to 1** (or *to 0*) if $I = (a, \infty)$ for $a \in \mathbb{R}$, the profile φ is monotonic, non-constant and $\varphi(\xi) \rightarrow 1$ (respectively, $\varphi(\xi) \rightarrow 0$) as $\xi \rightarrow \infty$; a **semi-wavefront from 1** (or *from 0*) if $I = (-\infty, b)$ for $b \in \mathbb{R}$, the profile φ is monotonic, non-constant and $\varphi(\xi) \rightarrow 1$ (respectively, $\varphi(\xi) \rightarrow 0$) as $\xi \rightarrow -\infty$.

Proposition 1

Assume that there exists a wavefront of (1) with profile φ satisfying (3), with speed c . Then, $\varphi'(\xi) < 0$ for every $\xi \in \mathbb{R}$ such that $\varphi(\xi) \neq 0, 1, \alpha, \gamma$. Moreover, we have:

- (i) When $\alpha \geq \gamma$, there exists a solution of (5).
- (ii) When $\alpha < \gamma$, there exists a solution of (6); if it holds

$$|g(\varphi)| \leq L|\varphi - \gamma| \text{ in a left- or right-neighborhood of } \gamma, \quad (7)$$

then $z(\gamma) > 0$ and hence z satisfies (5).

Moreover, when either $\alpha > \gamma$ or both $\alpha < \gamma$ and (7) hold, then there exists a unique $\xi_\gamma \in \mathbb{R}$ such that $\varphi(\xi_\gamma) = \gamma$, and it holds $\varphi'(\xi_\gamma) < 0$.

Proposition 2

For $c \in \mathbb{R}$, assume one of the following:

- (a) there exists a solution of (5) and either
 - (i) $\alpha \neq \gamma$, or
 - (ii) $\alpha = \gamma$, and $(\heartsuit)$ holds;
- (b) there exists a solution of (6) with $z(\gamma) = 0$, $\alpha < \gamma$, and $(\heartsuit)$ holds.

Then, Equation (1) admits a wavefront associated to c whose profile φ satisfies (3).

Corollary 1

We have:

- (a) *when $\alpha > \gamma$, there exists a one-to-one correspondence between profiles φ (up to space shifts) with speed c and solutions z of (5);*
- (b) *when $\alpha = \gamma$, under $(\heartsuit)$ there exists a one-to-one correspondence between profiles φ (up to space shifts and stretchings at γ) with speed c and solutions z of (5);*
- (c) *when $\alpha < \gamma$ and (7) holds, then there exists a one-to-one correspondence between profiles φ (up to space shifts) with speed c and solutions z of (5). Instead, under $(\heartsuit)$ there exists a one-to-one correspondence between profiles φ_0 of Theorem 1 (c) (up to shifts and stretchings at γ) and solutions of (6) with $z(\gamma) = 0$.*